

gpx-slope-colors

Colour your GPX tracks by slope — uphill in one colour, downhill in another — and see them that way in OsmAnd without a Pro subscription, and on Garmin too.

Thank you for downloading. Everything you need is in this folder.

The 30-second version

1. Double-click `gpx-slope-colors.exe`. A window opens.
2. Drag your `.gpx` files into it — or press *Add files*.
3. Press **Convert**.
4. A new file appears next to each original, named `yourtrack_slope.gpx`.
5. Copy it to your phone and open it with OsmAnd.

Uphill is red, downhill is green, flat is purple. All of it is adjustable in the window.

Try it first with `example/demo.gpx` to see what happens.

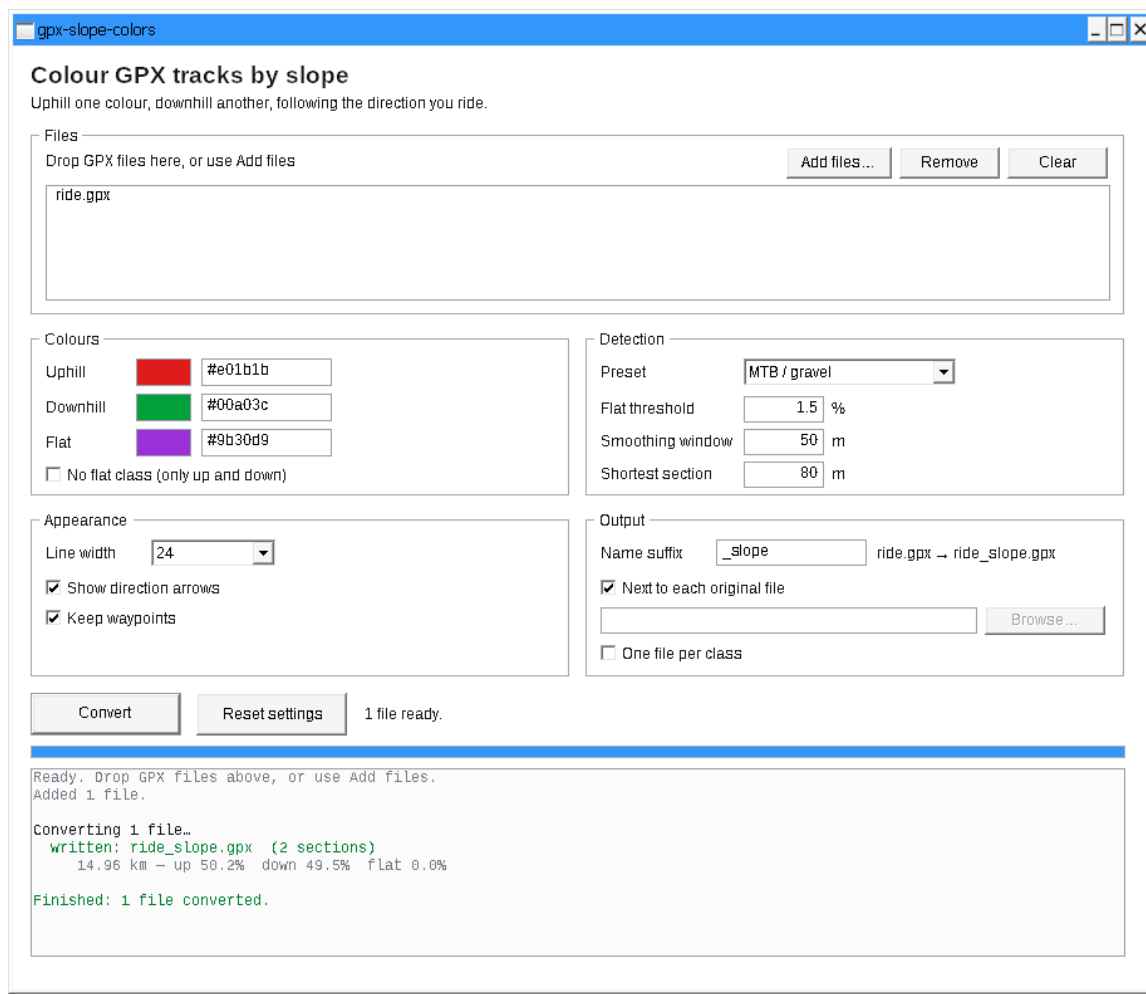
Nothing to install. No Python, no runtime, no setup program, no account. The `.exe` is a single self-contained file — you can run it from a USB stick if you like.

What's in this folder

File	What it is
<code>gpx-slope-colors.exe</code>	The app. Double-click this one
<code>gpx-slope-colors-cli.exe</code>	Command line version, for scripting and batch jobs
<code>example/demo.gpx</code>	A sample track, to try it out
<code>example/demo_slope.gpx</code>	What the sample looks like after processing
<code>source (mac and linux)/</code>	The full source code, and a one-line build script
<code>LICENSE</code>	MIT — you may use, modify and redistribute this freely

The `.exe` files are for **Windows (64-bit)**. On macOS and Linux you build from the included source in about five seconds — see *Mac and Linux* near the end.

Using the app



The app after converting two tracks.

Files — drag them in, or use *Add files*. Queue up as many tracks as you like and convert them in one go. Dropping a folder adds every GPX inside it. You can also drag files straight onto the **.exe** icon, or right-click a GPX file and choose *Open with*.

Colours — click a swatch to open a colour picker, or type a name (**orange**, **lightblue**) or a hex code straight into the box next to it. The swatch and the box stay in sync.

Detection — pick a preset for your activity, or set the three numbers yourself. Editing any of them switches the preset to *Custom*, which is just a label; nothing is lost.

Appearance — line width, direction arrows, and whether to carry over waypoints.

Output — the *Name suffix* box controls what gets added to the file name: **_slope** turns **ride.gpx** into **ride_slope.gpx**. Change it to anything you like. By default each result is written next to its original; untick that to send everything to one folder instead.

The log at the bottom shows what happened to each file, with the length and the uphill/downhill/flat split. A file that fails is reported in red with the reason, and the others still get processed.

Importing into OsmAnd

1. Get the file onto your phone — email it to yourself, use a cloud drive, or a USB cable.

2. Tap it and choose to open it with OsmAnd. Or, inside OsmAnd: **Menu** → **My Places** → **Tracks** → **Import**.
3. Show the track on the map.

Nothing to set under Appearance. The colours are already inside the file.

Needs OsmAnd **4.9.10 or newer**. Older versions collapse multi-track files to a single colour on import.

A nice side effect: the direction arrows along the track stay visible. They disappear when you use OsmAnd's own gradient colouring modes.

Does it work outside OsmAnd? Yes — on Garmin too

The file is **standard GPX 1.1** — it validates against the official schema — so any app, website or GPS device that reads GPX will open it. The colours show up in two places: **OsmAnd** and **Garmin**. What changes from one app to the next is only whether it shows the *colours*.

Where	Opens the file	Shows the colours
OsmAnd 4.9.10+ (free)	yes	yes , full palette
Garmin BaseCamp	yes	yes , nearest of 17 fixed colours
Garmin devices (Edge, Oregon, GPSMAP...)	yes	yes , nearest of 17 fixed colours
Strava, Komoot, Ride with GPS	yes	no — one colour
Google Earth, QGIS, Gaia, Locus	yes	no — one colour

GPX never standardised track colours, so every app invented its own tag. This file carries both of the tags that matter, which costs nothing and covers two families of device:

OsmAnd reads `<osmand:color>`, which takes any colour you like — you get exactly the shade you picked, plus the line width and the direction arrows.

Garmin reads `<gpxx:DisplayColor>`, and here there is a limit worth knowing: Garmin allows only **17 fixed colour names**, no hex codes. Your colour is matched to the closest one it permits — the default red stays **Red**, the green becomes **DarkGreen**, the purple **DarkMagenta**. The match is done by perceived colour, so it never lands in the wrong family. The tag is written **twice**, under both the `gpxx` and `gpxtrx` prefixes, because Garmin's own desktop software and its handhelds each read one and ignore the other — a file written by one Garmin product can otherwise lose its colours in another.

Every file is checked against Garmin's own published schema before release, and every colour name it writes is one the schema permits. Worth being straight with you: that is a guarantee about the file being correct, not a test on a physical device — we don't have a lab of Garmin units to try it on.

Apps that understand neither tag simply **ignore them** and draw the track in their own colour. Nothing breaks: points, elevations, timestamps, heart rate and cadence are all still there.

Strava and Komoot: don't convert, just upload the original

Those sites cannot show per-section colours at all, and no file can make them. A GPX track carries one colour; the colours here work because the ride is cut into several tracks, which is exactly what Strava and Komoot refuse to treat as one activity.

So there is nothing to gain by converting a file for them. **Upload your original .gpx instead** — the tool never modifies it — and keep the coloured copy for OsmAnd.

An earlier version of this tool had a "single track" option meant for those sites. It was removed, because the file it produced was the same ride, with the same points and the same sensor data, as the file you already had. It converted something into itself. If a feature only makes a copy, it shouldn't exist.

If the result looks too choppy

On long tracks the defaults produce a lot of short sections. Two settings control this — *Shortest section* and *Smoothing window* in the app, `--minlen` and `--window` on the command line.

Track length	Try
up to 20 km	the defaults
20–50 km	window 80 , shortest section 150
over 50 km	window 100 , shortest section 250

Shortest section is the one to turn first: it is the shortest stretch, in metres, that gets its own colour. Anything shorter is merged into its neighbour.

There is also the **flat threshold**, the gradient percentage below which ground counts as flat. Raise it if too much gentle terrain is being called a climb:

Activity	Suggested
Road cycling	1
MTB / gravel	1.5 (default)
Hiking	3

The presets in the app set all three at once.

Using it from a terminal

Everything the app does is also available on the command line, which is handy for scripting or batch jobs. Open Command Prompt in this folder and run:

```
gpx-slope-colors-cli.exe yourtrack.gpx --uphill orange --downhill lightblue
```

Some examples:

```
gpx-slope-colors-cli.exe ride.gpx
gpx-slope-colors-cli.exe ride.gpx -o coloured.gpx
gpx-slope-colors-cli.exe ride.gpx --suffix _osmand
gpx-slope-colors-cli.exe ride.gpx --window 100 --minlen 250 --width 16
gpx-slope-colors-cli.exe --output-dir coloured *.gpx
```

Available colour names:

red · darkred · orange · yellow · green · lightgreen · darkgreen · lightblue · blue · darkblue · cyan · purple · magenta · pink · brown · gray · lightgray · black · white

You can also use hex codes: `--uphill "#ff8800"`. To see the full list with the hex values, run `gpx-slope-colors-cli.exe --colors`.

All the options

Run `gpx-slope-colors-cli.exe --help` for the complete list with worked examples.

Option	Default	What it does
<code>-o, --output</code>	<code><name>_slope.gpx</code>	Output file name
<code>--output-dir DIR</code>	—	Write every result into one folder
<code>--suffix TEXT</code>	<code>_slope</code>	What gets added to the file name
<code>--threshold PCT</code>	<code>1.5</code>	Gradient % below which ground counts as flat
<code>--window M</code>	<code>50</code>	Metres used to smooth the elevation profile
<code>--minlen M</code>	<code>80</code>	Shortest section that gets its own colour
<code>--uphill COLOR</code>	<code>red</code>	Uphill colour
<code>--downhill COLOR</code>	<code>green</code>	Downhill colour
<code>--flat COLOR</code>	<code>purple</code>	Flat colour
<code>--no-flat</code>	<code>off</code>	Only two colours: uphill and downhill
<code>--split-files</code>	<code>off</code>	One file per class instead of one combined file
<code>--width W</code>	<code>24</code>	Line width: <code>thin</code> , <code>medium</code> , <code>bold</code> , or <code>1–24</code>
<code>--no-arrows</code>	<code>off</code>	Hide the direction arrows
<code>--keep-waypoints</code>	<code>off</code>	Carry the waypoints over to the new file
<code>--quiet</code>	<code>off</code>	Print errors only
<code>--colors</code>	—	List the colour names
<code>--version</code>	—	Print the version

You can pass several files at once, and wildcards work: `gpx-slope-colors-cli.exe *.gpx`.

Mac and Linux

The source code is in `source (mac and linux)/`, and building it takes seconds. Open a terminal in that folder and run:

```
sh build.sh
```

That produces an executable called `gpx-slope-colors` right there. Try it with:

```
./gpx-slope-colors ../example/demo.gpx
```

The only thing you need is a C++ compiler. On macOS, run `xcode-select --install` if you haven't already. On Debian or Ubuntu, `sudo apt install g++`. On Fedora, `sudo dnf install gcc-c++`. There are no libraries to fetch — the program depends on nothing but the standard library.

To use it from anywhere, copy it onto your PATH:

```
sudo cp gpx-slope-colors /usr/local/bin/
```

The graphical window is Windows-only; on macOS and Linux you get the command line version, which does everything the window does.

Troubleshooting

Windows says "Windows protected your PC" — click *More info*, then *Run anyway*. This is SmartScreen, and it appears for any program that hasn't been downloaded thousands of times or signed with a certificate that costs a few hundred euro a year. Nothing is wrong with the file. If you'd rather not take my word for it, the full source is in the folder and you can build it yourself.

Your antivirus flags it — same story, and it is a known nuisance with small, self-contained executables from unknown publishers. The program makes no network connections whatsoever; you can verify that with any firewall. Again, the source is right there.

"no elevation data (<ele>) in the GPX file" — the track has no altitude recorded, so there is no way to know what is uphill. Fix it in OsmAnd: open the track → **Analyse on map** → **Correct altitude**, then export and try again. Most online route planners can also add elevation to a GPX.

The colours don't show in OsmAnd — check you are on OsmAnd 4.9.10 or newer. On older versions, use `--split-files` and import the resulting files separately.

The track looks like a rainbow of tiny segments — raise the shortest section setting. See the section above.

The line is too thick — the default is maximum width. Try `16`.

My waypoints disappeared — tick *Keep waypoints*, or add `--keep-waypoints`.

The app won't start on an old PC — the `.exe` is 64-bit. 32-bit Windows is not supported; build from source if you need it.

Good to know

- Your files never leave your computer. The tool runs locally, with no network access and no telemetry.
 - The output is standard GPX 1.1. It opens in any app; apps that don't read the OsmAnd colour tags simply show the track in one colour.
 - Elevations, timestamps and per-point sensor data (heart rate, cadence, power, temperature) are all carried over untouched.
 - The tool never modifies your original file. It always writes a new one.
 - An existing output file is overwritten without asking.
-

Licence

MIT — see [LICENSE](#). In short: use it, change it, share it, build on it, including commercially. No warranty.

The full source is included in this download, and is also on GitHub if you want to follow changes or report a problem.

If this saved you the price of a Pro subscription, or just made your rides easier to read, consider telling someone about it. That helps more than anything.